Self-Repair as a Bounded Disturbance: Input-to-State Stability of Neural Network Training Dynamics

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Why this matters now

Networks that train for long lose plasticity: units die and stop learning (Dohare et al., *Nature* 2024). The field's fix is to repair them mid-training: Continual Backprop, ReDo (ICML 2023), Shrink & Perturb (NeurIPS 2020), plasticity injection (NeurIPS 2023). A repair is *not the gradient of any loss*, so the usual convergence proofs cover none of them.

When does a network that fixes itself still converge?

What's new: repair modelled as an input to the training dynamics; a stability bound for it; and a matching impossibility result that explains why resets are needed.

NETWORKS ARE NOW REPAIRED MID-TRAINING. WHEN IS THAT SAFE?

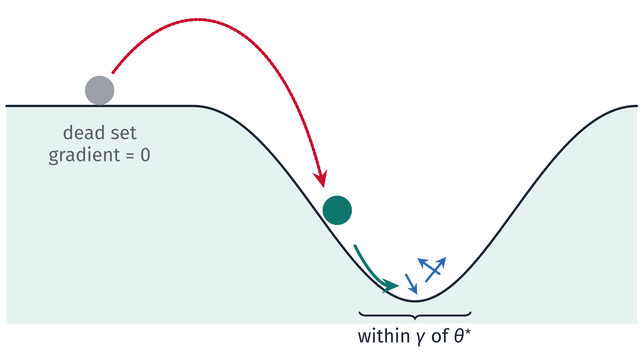
Small repairs keep training stable. Only a jump can revive a dead layer.

We prove both, which makes self-repair a switched system: bounded flicks between isolated jumps.

Repair as a control input

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t) + u_t$$

Treat each repair u_t as an *input* and ask for input-to-state stability (Sontag, 1989). Training becomes a switched system:



flow: the gradient step, contracting near θ^*
flicks: small capped corrections, $\|u_t\| \le \varepsilon_{max}$
jumps: resets that redraw a dead layer

The repair rule we analyse

Each layer gets a health score from three signals: the global loss, its own local loss through a shared readout, and its gradient stability. An alarm fires when health falls 2.5σ below its own running mean. Corrections are mean-centred and clipped to $0.9\|w\|$; a layer dead for three epochs is redrawn.

383 runs: 50-node DAG, two CNNs, ResNet-18; CIFAR-10/100.

THEOREM 1 · PRACTICAL ISS

Bounded repair is safe

Near a minimum θ^*, let L be L-smooth and α-strongly convex, $\eta \le 1/L$, repairs $\|u_t\| \le \varepsilon_{max}$, and local gradients (weight β) bounded by M. Then

$$\limsup_{t\to\infty} \|\theta_t - \theta^*\| \le \gamma = \frac{2\beta M}{\alpha} + \frac{\varepsilon_{max}}{\eta\alpha}$$

Repairs push training around, but never further than γ from a good solution.

THEOREM 2 · ABSORBING DEAD SET

Bounded repair cannot revive

In a Conv–BN–ReLU layer (BatchNorm in train mode) let channel c have zero filter W_c and BN shift $\beta_c \le 0$. This dead set D is invariant under AdamW (fresh state) and under every repair with $\|R(W) - W\| \le \rho\|W\|$:

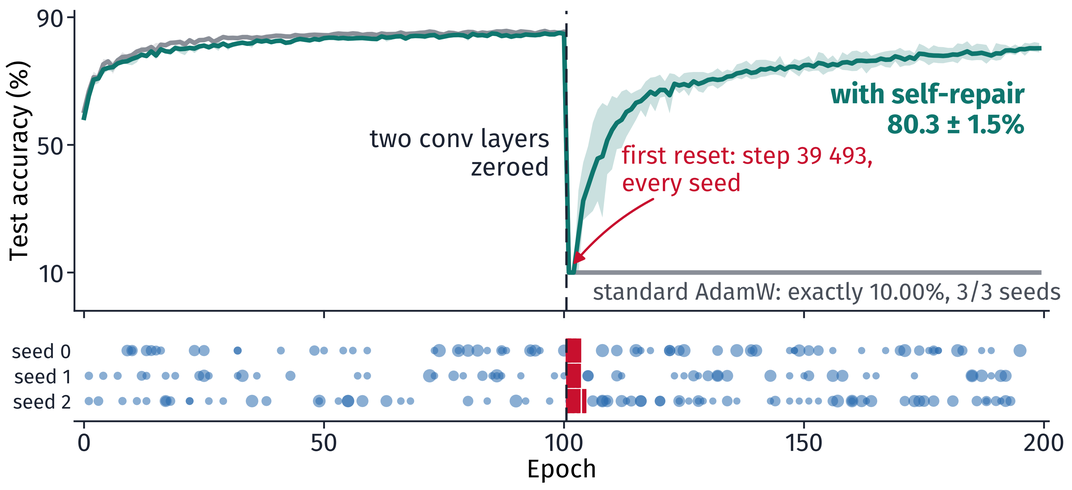
$$\theta_0 \in D \implies \theta_t \in D \ \ \forall t$$

Gradient descent and small fixes are both stuck, exactly. Only a jump leaves D.

10.00%
without repair, after the lesion: exactly chance, 99 epochs, 3/3 seeds

39 493
the step where the first reset fires, identical in every seed

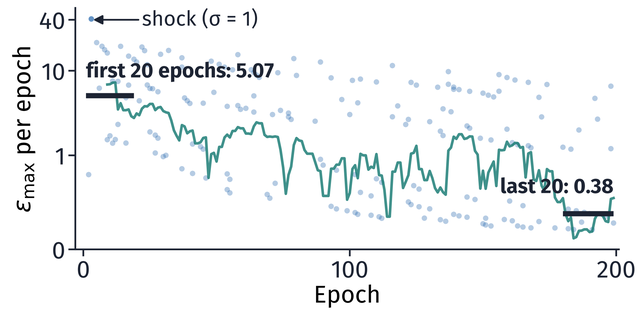
80.3%
with self-repair (85% before the lesion)



Theorem 2, in data. CIFAR-10 CNN; two conv layers zeroed after epoch 100. Bottom: every repair event, **small corrections** and **resets**.

A falsifiable prediction

Theorem 1's bound is useful only if repairs *shrink*. They do: after a shock, ε_{max} falls **5.07 → 0.38 (−93%)**.



Where repair stops mattering

On ResNet-18 the skip path routes around the same lesion, and the network heals unaided: **92.3%** without repair vs **91.6%** with (3 seeds). Repair matters when there is no redundancy to fall back on.

Proof ideas

Thm 1. The gradient step contracts by $\rho = 1 - \eta\alpha$, so $r_{k+1} \le \rho\, r_k + 2\eta\beta M + \varepsilon_k$; sum the geometric series.

Thm 2. A zero filter feeds BatchNorm a constant, so $\hat{x} \equiv 0$. With $\beta_c \le 0$ the ReLU is closed and every gradient into the channel is exactly 0. AdamW then gives $\beta_c(t) = (1 - \eta\lambda)^t \beta_c(0)$, which we see to the bit in the logs.

OPEN PROBLEM

How often may a network jump?

Theorem 1 holds between jumps; Theorem 2 says jumps are unavoidable. Find a dwell time τ_D such that

$$\text{jumps} \ge \tau_D \text{ apart} \implies \limsup_t r_t \le \gamma'$$

(cf. Hespanha & Morse, 1999). Also open: Polyak–Łojasiewicz in place of strong convexity, and $\varepsilon(t)$ derived rather than measured.

Scan: 3-min video, paper, ask the author

The author could not travel. Open to research collaborations; questions welcome:

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